

KISAN PRAVAH

Technical Architecture & Technology Overview

AI-enabled, multilingual procurement coordination platform connecting farmers and procurement centres from crop planning to payment.

Layer / Domain	Technology	Purpose
Frontend	Angular, TypeScript, Tailwind CSS, Progressive Web App	Farmer and procurement-centre interfaces with accessible, responsive workflows.
Client State & Data	Angular services / RxJS, form validation	Application state, registration, booking and dashboard synchronization.
Backend API	Node.js, Express.js, REST APIs, Socket.IO	Business logic, authentication, registration, recommendations, booking, tracking and real-time updates.
Database	MongoDB, Mongoose	Farmer profiles, crop declarations, centre details, slots, tokens, procurement and transaction status.
Real-Time Cache	Redis	Live queue positions, active tokens, centre load and fast temporary state.
AI & Prediction	Python, Scikit-learn, OpenAI API	Demand forecasting, arrival/quantity prediction, congestion/waiting-time prediction and multilingual assistance.
Integrations	Weather & MSP APIs, Google Maps/GPS	Crop-planning insights, centre recommendation and location services.
Communication	SMS Gateway, WhatsApp API, Firebase Cloud Messaging	Token, slot, queue, reminder, procurement and payment notifications.
Cloud & DevOps	GCP: Cloud Run, Cloud Storage, Vertex AI, Firebase Cloud Messaging, Cloud Monitoring; Docker, Git, GitHub, CI/CD	Deployment, storage, AI services, notifications, monitoring and delivery.
Security	OTP Authentication, RBAC, HTTPS/TLS, Data Encryption	Secure accounts, encrypted communication and role-specific access.

KISAN PRAVAH — END-TO-END ARCHITECTURE

Core workflow adapted from the submitted Kisan PravaH solution and technical approach.

FARMER PWA	CENTRE OPERATOR	GOVERNMENT / ADMIN
Crop + quantity declaration	Capacity & workload view	District monitoring
Registration + documents	Expected crop forecast	Demand & congestion insights
MSP / demand / weather insights	Queue & slot management	Procurement progress
Centre recommendation	Quality / weighing workflow	Centre utilisation
Slot / token booking	Procurement & payment updates	Planning & resources
Live queue + alerts	Real-time dashboard	Reports & analytics

BACKEND & AI FLOW

- 1. Interaction:** Angular PWA/dashboard receives farmer and centre requests; SMS and WhatsApp deliver low-friction alerts.
- 2. API & Database:** Node.js/Express handles business logic and Mongoose persists farmer, crop, centre, booking, procurement and payment data in MongoDB.
- 3. Real-Time:** Socket.IO updates queue/token status while Redis provides fast live-state access.
- 4. Prediction:** Python/Scikit-learn processes operational data for demand, arrival, quantity, congestion and waiting-time prediction.
- 5. Multilingual AI:** OpenAI API supports multilingual assistance around farmer-facing information and predictive outputs.
- 6. Communication:** SMS Gateway, WhatsApp API and Firebase Cloud Messaging send booking, queue, reminder, procurement and payment updates.
- 7. External Data:** Weather, MSP and Maps/GPS services support informed planning and centre recommendation.
- 8. Deployment:** Dockerized components are deployed on Google Cloud Platform with monitoring, Git/GitHub and CI/CD.

KISAN PRAVAH — IMPLEMENTATION FLOW

DECLARE → PREDICT → ALLOCATE → ARRIVE → PROCURE → TRACK

DECLARE: Farmer pre-registers crop and expected quantity.

PREDICT: AI forecasts expected arrivals, quantities, demand and workload.

ALLOCATE: Centre recommendation combines availability with MSP, demand, weather and location insights; farmer books slot/token.

ARRIVE: Alerts communicate slot, documents, quality guidance and queue status.

PROCURE: Centre manages arrival, quality, weighing, storage and procurement workflow.

TRACK: Farmer follows procurement and payment status digitally.

METHODOLOGY

Analyze → Prototype → Develop → Integrate → Pilot Test → Deploy on GCP → Improve

SECURITY & SCALABILITY

OTP authentication, role-based access, HTTPS/TLS and data encryption are used. The modular cloud architecture is designed to expand from pilot centres to wider procurement networks.